

5.7 Interpreting coefficients

A fitted regression is usually reported as a table of coefficients. The short answer, “the change in the mean response per unit change in the regressor, holding the other regressors fixed”, is a useful slogan and a frequent source of error. This section makes the slogan precise within the model, shows how the meaning of a coefficient depends on units, centring, transformations and the other columns of the model, and marks where the slogan stops being true.

5.7.1 What a coefficient measures

Definition 5.7.1. Interpretation of a coefficient

In a linear model $E(Y | \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$, suppose the regressor x_j enters the model only through its own column, so that no other column is a function of x_j . For any two regressor profiles $\mathbf{x}$ and $\mathbf{x}'$ that agree in every regressor except the j th, and have $x'_j = x_j + 1$,

$$E(Y | \mathbf{x}') - E(Y | \mathbf{x}) = \beta_j. \quad (5.7.1)$$

The coefficient β_j is the **partial effect** of x_j in this model: the difference in mean response per unit difference in x_j , between profiles that agree on all the other regressors in the model. Its estimate $\hat{\beta}_j$ is the least squares estimate of this difference.

Three qualifications are built into the definition.

1. **It compares profiles, not interventions.** Eq. 5.7.1 compares the mean response of two kinds of case that exist in the population the model describes. Whether changing x_j in a given case would change its response by β_j is a causal question, which the model alone cannot answer (Chapter 25).
2. **“The other regressors” means the ones in the model.** Adding or removing a regressor changes the comparison being made, and so changes β_j , unless the regressors are orthogonal. Proposition 5.7.2 gives the exact relation.
3. **It needs x_j to vary alone.** If x_j also appears in other columns, as in a polynomial or an interaction, there are no two profiles that differ only in the x_j column, and β_j alone does not measure the effect of x_j .

The geometric content of “holding the others fixed” in a data set, where nobody held anything fixed, is the Frisch–Waugh–Lovell theorem of Section 6.6: $\hat{\beta}_j$ is the slope of the response on the part of x_j that is not linearly explained by the other regressors.

5.7.2 Units, rescaling and reparameterization

The coefficient β_j is measured in units of the response per unit of x_j . Changing units rescales it. More generally, any invertible linear recoding of the columns changes the coefficients in a predictable way and leaves the fit untouched.

Proposition 5.7.1. Reparameterization

Let $\mathbf{X}$ be $n \times p$ of rank p , let $\mathbf{K}$ be a nonsingular $p \times p$ matrix, and put $\mathbf{Z} = \mathbf{X}\mathbf{K}$. Then $\mathbf{Z}$

has rank p , and the least squares estimates from the two model matrices are related by

$$\hat{\boldsymbol{\beta}}_Z = \mathbf{K}^{-1} \hat{\boldsymbol{\beta}}_X.$$

The fitted values, residuals, SSE and s^2 are the same for both, and the estimated covariance matrices satisfy $s^2(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1}[s^2(\mathbf{X}^\top \mathbf{X})^{-1}](\mathbf{K}^{-1})^\top$.

Proof. $\text{rank}(\mathbf{Z}) = \text{rank}(\mathbf{X}) = p$ by Proposition 1.2.3(b). Then

$$\begin{aligned} \hat{\boldsymbol{\beta}}_Z &= (\mathbf{K}^\top \mathbf{X}^\top \mathbf{X} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^\top)^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{K}^{-1} \hat{\boldsymbol{\beta}}_X, \end{aligned}$$

and $\mathbf{Z} \hat{\boldsymbol{\beta}}_Z = \mathbf{X} \mathbf{K} \mathbf{K}^{-1} \hat{\boldsymbol{\beta}}_X = \mathbf{X} \hat{\boldsymbol{\beta}}_X$. Equal fitted values give equal residuals, SSE and s^2 . The covariance identity follows from $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^{-1})^\top$. ■

With $\mathbf{K} = \text{diag}(1, c_1, \dots, c_k)$, measuring x_j in units $1/c_j$ times as large divides its coefficient and its standard error by c_j , and leaves everything else alone. Centring, effect coding of a factor (Exercise (B1)) and many other changes of coding are also of the form $\mathbf{X} \mathbf{K}$. Section 6.7 shows that what really stays fixed is the column space of $\mathbf{X}$.

The intercept and centring. The intercept β_0 is the mean response at the profile where every regressor is zero. That profile may be impossible (zero income) or far outside the data (an August sea temperature of 0°C in Example (Forecasting December sea temperature)). If the regressors are centred, the intercept becomes the mean response at the average profile $\bar{\mathbf{x}}$, and its estimate is $\bar{y}$ (Proposition 5.4.3). Centring does not change the slopes. It makes the intercept interpretable, and it makes $\hat{\beta}_0$ uncorrelated with the slopes (Exercise (B2)). With interactions or polynomials, centring also changes what the lower-order coefficients mean, as the next subsections show.

Standardized coefficients. Measuring every variable in units of its own sample standard deviation gives the **standardized coefficients** $\hat{\beta}_j^* = \hat{\beta}_j s_{x_j}/s_y$: the difference in the response, in standard deviations, per standard deviation of x_j . They make coefficients unit-free and are sometimes used to rank the “importance” of regressors. That use is fragile. A standard deviation is a property of the sample, not of the relation: the same relation observed over a wider range of x_j gives a larger standardized coefficient.

5.7.3 Logarithms

When the response or a regressor is on a logarithmic scale, the coefficients describe proportional rather than absolute differences.

- **Log–log**, $E(\log Y) = \beta_0 + \beta_1 \log x + \dots$: a 1 percent difference in x goes with approximately a β_1 percent difference in Y (exactly, a factor 1.01^{β_1} in the geometric mean). Economists call β_1 an **elasticity**.
- **Log–level**, $E(\log Y) = \beta_0 + \beta_1 x + \dots$: a unit difference in x goes with a factor e^{β_1} in the geometric mean of Y , a difference of $100(e^{\beta_1} - 1)$ percent, approximately $100\beta_1$ percent when β_1 is small.
- **Level–log**, $E(Y) = \beta_0 + \beta_1 \log x + \dots$: a 1 percent difference in x goes with a difference

of about $\beta_1/100$ in Y .

The phrase “geometric mean” matters. $E(\log Y)$ is not $\log E(Y)$, so a model for the mean of $\log Y$ is not a model for the mean of Y . [Chapter 22](#) returns to back-transformation (Proposition 22.5.1).

Example. Reading the money demand coefficients

In Example (Money demand in Denmark) the response is log real money and the first regressor is log real income, so $\hat{\beta}_1 = 1.2958$ is an income elasticity. Comparing quarters with the same interest rates, 1 percent more real income goes with about 1.3 percent more real money. The interest rates are measured as fractions, so the bond rate coefficient -2.6163 is per unit of the fraction, that is, per 100 percentage points, an absurdly large change. Rescaling the rate to percentage points ([Proposition 5.7.1](#) with $c = 100$) divides the coefficient by 100: a bond rate one percentage point higher, with income and the deposit rate the same, goes with money holdings lower by the factor $e^{-0.026163}$, a difference of -2.58 percent. The deposit rate has a positive coefficient, which is plausible if deposits are part of the money measured, since a higher deposit rate makes holding them more attractive. But the coefficient is smaller than its standard error.

The standardized coefficients of income, bond rate and deposit rate are 0.619, -0.529 and 0.057. The sample standard deviation of log income is 0.0728, about 7 percent, and that of the bond rate is 0.0308, about 3 percentage points; within those ranges of variation, income and the bond rate move money holdings by similar amounts.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve  $X^T X b = X^T y$ 

X_pct = X.copy()
X_pct[:, 2:] *= 100                          # rates in percentage points
beta_pct = np.linalg.solve(X_pct.T @ X_pct, X_pct.T @ y)
print("rates as fractions:", np.round(beta_hat, 4))
print("rates in percent:  ", np.round(beta_pct, 5))
print("same fitted values:", np.allclose(X @ beta_hat, X_pct @ beta_pct))

sd = X[:, 1:].std(axis=0, ddof=1)
standardized = beta_hat[1:] * sd / y.std(ddof=1)
print("standardized slopes:", np.round(standardized, 3))
```

[Run this code online](#)

5.7.4 Terms that move together

If x enters through both x and x^2 , the mean response $\beta_0 + \beta_1 x + \beta_2 x^2$ changes at the rate

$$\frac{\partial E(Y | x)}{\partial x} = \beta_1 + 2\beta_2 x,$$

which depends on x . The coefficient β_1 alone is this rate at $x = 0$, and β_2 is half the rate at which the slope itself changes. Neither is “the effect of x ”. In Example (A curved trend and a seasonal cycle) the growth rate of carbon dioxide was 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The coefficient $\hat{\beta}_1 = 1.3344$ is the growth rate in 1980, the year chosen as the origin of t_c , and would have been different had the origin been chosen differently.

Interactions behave the same way. In Eq. 5.3.3 the slope in x_1 is $\beta_1 + \beta_3 x_2$. Reporting “the effect of x_1 ” requires choosing a value of x_2 , and β_1 is the choice $x_2 = 0$. Centring x_2 at a representative value before forming the product makes β_1 the slope at that value. The fitted surface does not change (Exercise (B2)).

5.7.5 Indicator variables

With reference coding, the coefficient of an indicator is the difference in mean response between its category and the reference category, among profiles that agree on the other regressors. Changing the reference changes every indicator coefficient but not the fit, since the reference coding for one reference category is a reparameterization of the coding for any other.

Example. Self-rated health, with and without adjustment

In Example (Self-rated health), people reporting poor health made 3.160 more physician visits a year, on average, than those reporting excellent health. Add two regressors: a chronic-disease score (the variable `disea`) and an indicator of a physical limitation. The coefficient of poor health falls to 1.554, and that of fair health from 1.058 to 0.202. The good-health coefficient changes sign, to -0.081 . The adjusted comparison is between people with the same disease score and limitation status. Much of the difference in visits between health categories goes with differences in measured disease, and the self-rating adds less once disease is taken into account. Neither set of numbers is “the” effect of self-rated health. They answer different questions.

The data record the same people in several years, so the cases are not independent and the second-moment assumptions do not hold exactly; [Chapter 33](#) treats such clustered data.

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float)  # good, fair, poor
X = np.column_stack([np.ones(len(y)), D])                  # excellent =
    reference

Xa = np.column_stack([X, rand["disea"], rand["physlm"]])
beta_a = np.linalg.solve(Xa.T @ Xa, Xa.T @ y)
print("adjusted for chronic-disease score and physical limitation:")
print(np.round(beta_a, 4))
```

[Run this code online](#)

5.7.6 Omitted regressors

How does a coefficient change when other regressors are added or removed? Partition the model matrix as $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, with $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ the estimate from the full (“long”) regression, and

let $\tilde{\beta}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y}$ be the estimate from the “short” regression on $\mathbf{X}_1$ alone.

Proposition 5.7.2. Omitted regressors

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full rank, and let $\mathbf{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$ be the coefficient matrix from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$. Then:

- (a) for every data vector $\mathbf{y}$, $\tilde{\beta}_1 = \hat{\beta}_1 + \mathbf{\Pi} \hat{\beta}_2$;
- (b) if $E(\mathbf{Y}) = \mathbf{X}_1 \beta_1 + \mathbf{X}_2 \beta_2$, then $E(\tilde{\beta}_1) = \beta_1 + \mathbf{\Pi} \beta_2$.

In particular the short and long coefficients agree, for all $\mathbf{y}$, iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.

Proof. (a) Write $\mathbf{y} = \mathbf{X}_1 \hat{\beta}_1 + \mathbf{X}_2 \hat{\beta}_2 + \hat{\epsilon}$, where $\mathbf{X}^\top \hat{\epsilon} = \mathbf{0}$ and so $\mathbf{X}_1^\top \hat{\epsilon} = \mathbf{0}$. Then

$$\begin{aligned} \tilde{\beta}_1 &= (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y} \\ &= \hat{\beta}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \hat{\beta}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \hat{\epsilon} = \hat{\beta}_1 + \mathbf{\Pi} \hat{\beta}_2. \end{aligned}$$

(b) Take expectations in (a), using $E(\hat{\beta}) = \beta$ (Theorem 5.5.1), or apply Proposition 5.5.7(a) with $\delta = \mathbf{X}_2 \beta_2$. For the last claim, if $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ then $\mathbf{\Pi} = \mathbf{0}$. Conversely, if $\mathbf{\Pi} \neq \mathbf{0}$, take $\mathbf{y} = \mathbf{X}_2 \mathbf{c}$ with $\mathbf{\Pi} \mathbf{c} \neq \mathbf{0}$: the long regression gives $\hat{\beta}_2 = \mathbf{c}$, and the two coefficients differ by $\mathbf{\Pi} \mathbf{c}$. ■

Part (b) is the **omitted-variable bias** formula. The short regression does not estimate β_1 but $\beta_1 + \mathbf{\Pi} \beta_2$. Each omitted regressor contributes its own coefficient times its association with the included ones. The bias is zero if the omitted regressors have no effect or are uncorrelated with the included ones, in the sense $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$; otherwise it is there, and it can have either sign. Part (a) says the same thing about the numbers actually computed from a data set, with no model assumptions at all.

Example. Income without interest rates

Regressing log money on log income alone, with an intercept, gives an income elasticity of 1.819, against 1.2958 when both interest rates are included. Proposition 5.7.2(a) accounts for the whole difference. Regressing each rate on log income gives slopes -0.2215 (bond rate) and -0.0917 (deposit rate): in these data, quarters with higher income had lower interest rates, and the correlation of log income with the bond rate is -0.52 . The short-regression elasticity therefore absorbs

$$(-0.2215)(-2.6163) + (-0.0917)(0.6186) = 0.5229,$$

mostly the association of higher income with lower bond rates, which in turn go with more money. Which elasticity is the “right” one depends on the question. The long regression compares quarters with equal interest rates. The short one compares quarters as they came, with whatever rates accompanied a given income.

```
X1 = X[:, :2]                # intercept and log income
X2 = X[:, 2:]               # the two interest rates
b_short = np.linalg.solve(X1.T @ X1, X1.T @ y)
```

```
A = np.linalg.solve(X1.T @ X1, X1.T @ X2)    # regress each rate on (1, lry)
print("short regression slope on income:", round(b_short[1], 4))
print("long slope + A @ (rate coefs):    ", round(beta_hat[1] + A[1] @ beta_hat[2:],
      4))
print("auxiliary slopes of the rates on income:", np.round(A[1], 4))
```

[Run this code online](#)

Warning (Adjustment is not a cure). [Proposition 5.7.2](#) suggests a tempting strategy: include every variable that might be relevant, so that nothing is omitted. Two problems remain. First, some variables should *not* be held fixed. If a treatment works partly by changing an intermediate variable, adjusting for that variable removes part of the effect one wants to measure. Adjusting for a common consequence of the regressor and the response can create an association where there is none. Second, adding regressors that are nearly collinear with x_j inflates the variance of $\hat{\beta}_j$ ([Chapter 26](#), Definition 26.2.1). Which variables to adjust for is a question about how the data were generated, not about the fit. [Chapter 25](#) develops the tools for answering it.

5.7.7 Exercises

5.7.7.1 A. Check your understanding

Exercise. A1

In Example (Forecasting December sea temperature), suppose both temperatures are converted to degrees Fahrenheit, $F = 32 + 1.8C$. What happens to the slope, the intercept, R^2 and s ? Answer using [Proposition 5.7.1](#), noting that the response is also transformed.

5.7.7.2 B. Practice

Exercise. B1

Show that the standardized coefficients are the slopes obtained by regressing the standardized response $(y_i - \bar{y})/s_y$ on the standardized regressors $(x_{ij} - \bar{x}_j)/s_{x_j}$, without an intercept. For a single regressor, show that the standardized slope is the correlation r .

Solution. Let $\mathbf{X}_c$ be the centred regressors and $\mathbf{D} = \text{diag}(s_{x_1}, \dots, s_{x_k})$. The standardized regressors are $\mathbf{X}_c \mathbf{D}^{-1}$ and the standardized response is $(\mathbf{y} - \bar{y}\mathbf{1})/s_y$. By [Proposition 5.4.3](#) the slopes of $\mathbf{y}$ on $\mathbf{X}_c$ are $\hat{\beta}_1$. Dividing the response by s_y divides them by s_y , since least squares is linear in the response; replacing $\mathbf{X}_c$ by $\mathbf{X}_c \mathbf{D}^{-1}$ multiplies them by $\mathbf{D}$ ([Proposition 5.7.1](#) with $\mathbf{K} = \mathbf{D}^{-1}$). The result is $\mathbf{D}\hat{\beta}_1/s_y$, whose entries are $\hat{\beta}_j s_{x_j}/s_y$. For one regressor this is $(S_{xy}/S_{xx})\sqrt{S_{xx}/S_{yy}} = r$, as in [Eq. 5.2.6](#).

Exercise. B2

In the centred parameterization $E(\mathbf{Y}) = \alpha\mathbf{1} + \mathbf{X}_c\beta_1$ of [Proposition 5.4.3](#), show that $\text{Cov}(\hat{\alpha}, \hat{\beta}_1) =$

$\mathbf{0}$, $\text{Var}(\hat{\alpha}) = \sigma^2/n$ and $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1}$. Deduce that $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + \bar{\mathbf{x}}^\top (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}})$, which generalizes Corollary 5.5.4.

Solution. The matrix $[\mathbf{1}, \mathbf{X}_c]$ has full rank and $[\mathbf{1}, \mathbf{X}_c]^\top [\mathbf{1}, \mathbf{X}_c] = \text{diag}(n, \mathbf{X}_c^\top \mathbf{X}_c)$, because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$. By Theorem 5.5.1 the covariance of $(\hat{\alpha}, \hat{\boldsymbol{\beta}}_1^\top)^\top$ is $\sigma^2 \text{diag}(1/n, (\mathbf{X}_c^\top \mathbf{X}_c)^{-1})$, which gives all three claims. Since $\hat{\beta}_0 = \hat{\alpha} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1$, Corollary 5.5.2 gives $\text{Var}(\hat{\beta}_0) = \sigma^2/n + \bar{\mathbf{x}}^\top \sigma^2 (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}}$, the cross term being zero. With one regressor, $\mathbf{X}_c^\top \mathbf{X}_c = S_{xx}$.

Exercise. B3

In a model with an intercept and two regressors, let the short regression omit x_2 . Show that the bias of the slope on x_1 is $\beta_2 S_{12}/S_{11}$, where $S_{12} = \sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$ and $S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2$. Tabulate the sign of the bias according to the signs of β_2 and the correlation of the two regressors.

5.7.7.3 C. Going deeper

Exercise. C1

Construct a data set with $n = 8$ cases, two regressors and an intercept, in which the slope of y on x_1 is positive in the simple regression and negative in the regression on both x_1 and x_2 . Explain the reversal with Proposition 5.7.2, and draw the scatterplot of y against x_1 with the cases labelled by x_2 .